

Residual disability 10 years after falling ill in Guillain–Barré syndrome: A prospective follow-up study

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ABSTRACT

Objective: To describe residual disability 10 years after onset of Guillain–Barré syndrome (GBS) and longitudinal changes from 2 weeks after onset until 10 years afterwards. The Erasmus GBS Outcome score (EGOS) was applied for predicting prognosis at 2 and 10 years.

Methods: Twenty-nine patients, mean age at onset 49 years, were followed prospectively from 2 weeks to 10 years after GBS onset. Measures included; GBS disability score, EGOS, Barthel Index, Frenchay Activity Index, Sickness Impact Profile (SIP), Overall Neuropathy Limitations Scale (ONLS), Walk-12, and Fatigue Severity Scale.

Results: At 10 years, the facial paralysis found in 5 participants at 2 years was still present, 11 participants (38%) experienced paresthesia, 6 (21%) had limitations in their arms, and 15 (52%) had limitations in walking. Decreased health-related quality of life on comparison to the general population was seen in the physical dimension of SIP at 10 years. The median EGOS at 2 weeks was 4.5, which correlated highly only with the Barthel Index at 2 years and the ONLS arm scale at 10 years.

Conclusion: The residual disabilities at 1–2 years comprised mainly of reduced walking ability, and are still persistent 10 years after GBS onset. For some individuals, facial paralysis caused major disability. The EGOS only partly predicted residual disability at 2 and 10 years after onset.

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1. Introduction

Guillain–Barré syndrome (GBS) is an inflammatory demyelinating polyradiculoneuropathy of acute or sub acute onset that occurs in all parts of the world and at all ages [1]. GBS typically begins with paresthesia in the hands or feet, followed within hours to days by leg and arm weakness. The cardinal clinical features consist of progressive relatively symmetrical weakness, mild sensory symptoms and areflexia, but effects on the cranial and respiratory muscles as well as the autonomic nervous system are not uncommon [2].

We have previously published prospective studies on residual disability between the time-point of GBS onset and up to 2 years later in a population-based Swedish cohort (n=42) [3,4]. The primary recovery of function occurred during the first 6 months after onset and moderate to severe disability was seen in about 25% of the participants at 2 years after onset [3,4].

Other prospective clinical studies report similar overall proportions of residual disability at 1–2 years after onset, with moderate to severe disabilities seen in 10–31% of persons at 1 year after onset [5–7] and in 16–20% at 2 years [8,9]. Long-term functional status has so far only been studied in cross-sectional studies, which have shown residual signs in 31–48% of those examined at different time-points after onset and up to 6 years afterwards (10–13). A good outcome after GBS has mostly been associated with the capability to regain walking ability, which is often assessed with the GBS disability score [10,11]. Apart from physical disabilities, studies have shown decreased health-related quality of life (HRQL) and higher fatigue 3–8 years after GBS onset [12–14]. Subjective experiences of HRQL and fatigue also seem to differ between persons with residual symptoms and those who have fully recovered [12]. To our knowledge no prospective study has yet been published describing physical and psychosocial health status in persons with GBS at a later time-point than 2 years after onset.

Identifying the factors which predict a poorer outcome is important for planning of rehabilitation services, as well as providing adequate information on prognosis in the early phase after onset. The Erasmus GBS Outcome score (EGOS) has recently been suggested to predict outcome at 6 months after onset of GBS [15]. This scoring system is based on age

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at onset, preceding diarrhea and GBS disability score [10] at 2 weeks after onset. However, to our knowledge, it has not been tested for predicting outcome at a time-point later than 6 months after onset of GBS.

The aims of this study were: 1) to describe residual disability 10 years after GBS onset; 2) to describe longitudinal changes from 2 weeks to 10 years after GBS onset; 3) to compare self-rated perceptions of HRQL, fatigue and walking between persons with residual physical symptoms and those who have fully recovered at 10 years after onset of GBS; and 4) to apply the Erasmus GBS outcome score [15] for predicting outcome at 2 years and 10 years after onset of GBS.

2. Methods

2.1. Participants

The study group consisted of the same cohort that was previously followed prospectively for 2 years [3,4]. These persons were identified in 1998–99 from eight hospitals in central Sweden: Karolinska University Hospital at Huddinge and at Solna, Uppsala University Hospital, Örebro University Hospital, Västerås Hospital, Falun Hospital, Borås Hospital and Sahlgrenska University Hospital [3]. Of the 49 patients initially included, 42 were assessed at 2 weeks, 2 months, 6 months, 1 year and 2 years after onset of GBS.

At year 10 in 2009, 35 of the original 42 participants were still alive. An information letter was sent to all 35, and about a week later they were contacted by phone by the first author (AF) and invited to participate. Four of them declined participation and two could not be traced by phone or mail, and so the final study group included 29 participants.

2.2. GBS-specific measures

The GBS disability score (range 0–7) was applied at 2 weeks, 2 years and 10 years after onset [10,11]. A GBS disability score of 0 was judged as being fully recovered.

The GBS disability score at 2 weeks after GBS onset was used to retrospectively calculate the Erasmus GBS Outcome score (range 1–7) [15]. Older age at onset, preceding diarrhea and a higher GBS disability score at 2 weeks after GBS onset all result in a higher outcome score, predicting a worse outcome.

2.3. Measures performed at 2 weeks–10 years

Paresthesia was graded by the participants themselves as either absent or present. Facial function was tested using five facial movements that were graded as affected or not affected [3].

The Barthel Index (BI) [16] was used as a generic measure of activities of daily living (ADL). It comprises ten items including personal hygiene, indoors mobility and handling stairs. The total score ranges from 0 to 20 with lower scores indicating dependency in ADL.

The Frenchay Activities Index (FAI) [17] was also used; this is a self-rating measure containing 15 items which assess the frequency of doing chores both indoors and outdoors. Each item is scored from 0 to 3, with a higher score indicating that the person is more active and participates more in daily life.

HRQL was measured using the Swedish version [18] of the Sickness Impact Profile (SIP) [19]. The SIP is a generic self-rating instrument developed to measure the overall impact of disease. It consists of 12 categories with a total of 136 items that each describe an activity or aspect of behavior, or specify a dysfunction. Three of these categories form an aggregated physical dimension: Ambulation, Mobility and Body care and movement. Four other categories form an aggregated psychosocial dimension: Social Interaction, Alertness behavior, Emotional behavior and Communication. The remaining five categories are independent: Sleep and rest, Eating, Work, Home management and Recreation and pastimes. The score range for each

category is 0–100 with higher scores indicating higher perceived dysfunction.

2.4. Measures added at the 10-year follow-up

The Overall Neuropathy Limitations Scale (ONLS) [20] which is a modification of the Overall Disability Sum Score [21], was developed to assess the limitations of patients with immune-mediated peripheral neuropathies by measuring disability in the arms and legs. In the present study, ONLS scoring was performed only at 10 years after onset, since the scale was not available until 2006. Scores on the ONLS have been shown to correlate moderately with the physical subscale of the SF-36 and the time taken to walk 10 m [20]. The arm scale includes five items concerned with daily life, each of which is scored from 0 to 5, where 0 corresponds to normal function and 5 to the prevention of all purposeful movements. Items on the leg scale are scored from 0 to 7, where 0 corresponds to no effect on walking/running/climbing stairs and 7 to the prevention of all purposeful movements.

The 12-item Walking Scale (Walk-12) is a self-rating measure of walking ability. It was originally developed in English for persons with multiple sclerosis [22], and later translated to Swedish [23], and also modified for use in patients with peripheral neuropathy [24]. Participants use a response scale ranging from 1 (not at all limited) to 5 (extremely limited) to rate the extent to which their residual GBS symptoms limit their ability to walk, run, climb stairs, and perform standing activities. A transformed score of 0 to 100 is then calculated, with higher scores indicating a greater impact on walking [24].

The Fatigue Severity scale (FSS) [25] contains nine items that are scored from 1 to 7, with a higher sum score indicating more negative impact of fatigue on daily life. The FSS was originally developed for multiple sclerosis but has also been used in populations with immune-mediated polyneuropathy [26].

2.5. Procedure

The procedure of the 2-year follow-up study has previously been described [3,4]. The present study used data collected at 2 weeks, 1 year and 2 years after onset. The 10-year follow-up occurred at 10 years $\pm$ 6 months for 20, of the 29 participants, and 1–8 months after this time-point for the remaining nine. All data were collected by one of the authors (AF) during home visits. The self-rating scales were administered at the test occasion thus allowing the participants to ask questions if they wished. The study was approved by the Regional Ethics Committee of Karolinska Institutet in Stockholm, Sweden.

2.6. Statistical analysis

Medians, inter-quartile ranges (IQR), means and standard deviations (SD) of the data were calculated [27]. Non-parametrical statistical methods were used since the data were primarily ordinal. Changes in the presence of paresthesia were analyzed using a chi-square test. The Page test was used to analyze changes over time in the GBS disability score, the BI, the FAI and the SIP. The sign test was used to analyze changes between the different test occasions and to compare individual scores on the SIP at 10 years after GBS onset with age-matched Swedish reference data [28]. The Mann–Whitney *U*-test was used to analyze differences in age and GBS disability score at onset between participants who were followed-up at 10 years and those who were not, and in self-reported measures at 10 years between participants with residual symptoms (here defined as GBS disability score $\geq$ 1) and those who had fully recovered (GBS disability score = 0). The Spearman correlation coefficient was calculated between the EGOS, clinical factors and results of measures at 2 and 10 years after

GBS onset. A probability value of $p < 0.05$ was considered a statistical significant change. Data were analyzed using two software packages: SPSS (version 15.0) and StatXact-4.

3. Results

3.1. Background characteristics

The characteristics at onset of GBS of the initial study group ($n = 42$) and of the participants in the 10-year follow-up study ($n = 29$) are presented in Table 1. Pre-existing illnesses before GBS onset were mainly hypertension and diabetes mellitus. No participant had a pre-existing illness that restricted their daily activities. The 29 participants who were followed up at 10 years were significantly younger at onset (mean age 49 years) compared to the seven deceased persons (mean age 66 years), but not compared to the six who declined to participate or could not be traced (mean age 53 years). There were no significant differences between these subgroups on the GBS disability score at 2 weeks after onset. At 10 years after GBS onset, the mean age of the 29 participants was 59 years. Fifteen of them (52%) were working or studying, 13 (45%) had retired due to age and one (3%) had retired due to illness.

Most patients received intravenous immunoglobulin (IVIG) treatment because it was the preferred treatment at the different hospitals. The time between onset of symptoms and start of treatment, median 7 days (IQR 3–12), was not an indicator of outcome. Correlation coefficients were low between onset of symptoms to start of treatment and score on the GBS disability scale at 2 years ($r = 0.16$) and 10 years ($r = 0.16$).

3.2. Residual disability at 10 years after GBS onset and changes from 2 weeks to 10 years

GBS disability score decreased significantly between 2 weeks and 2 years (Table 2). Eight participants (28%) required ventilator support during the acute phase. Four participants, all of whom were among these eight, continued to improve slightly between 2 years and 10 years indicating a prolonged recovery.

The number of participants experiencing paresthesia decreased significantly over time: 27 participants (93%) at 2 weeks after GBS onset, 21 (72%) at 1 year, 17 (59%) at 2 years, and 11 participants (38%) at 10 years. On all occasions tingling was prominent in the feet and hands rather than in proximal areas of the extremities.

Facial function was affected in 13 participants (45%) at 2 weeks, 7 (24%) at 1 year, and 5 (17%) at 2, and at 10 years. The area of most

Table 1
Characteristics of the participants at 2 weeks after onset of Guillain–Barré syndrome in the initial prospective study and the 10-year follow up study.

	2-year follow-up study ($n = 42$)	10-year follow-up study ($n = 29$)
Age, mean (range)	52 (20–80)	49 (20–77)
GBS disability score, median (range)	4 (1–5)	4 (2–5)
Erasmus GBS Outcome score, median (range)	4.5 (1–6)	4.5 (2–6)
	Numbers (%)	Numbers (%)
Men/women	24 (57)/18 (43)	16 (55)/13 (45)
Dependent in activities of daily living before onset of GBS ^a	1 (2)	0
Antecedent medical events ^b	30 (71)	22 (76)
Pre-existing illness before GBS	16 (38)	10 (34)
Immunomodulatory treatment in acute phase of GBS	36 (86)	26 (90)
Ventilatory support in acute phase	9 (21)	8 (28)

^a Assessed with the Barthel Index.

^b Respiratory, gastric and other infections, surgery, vaccination, pregnancy in the four weeks before GBS onset.

Table 2

GBS disability score at 2 weeks, 2 years, and 10 years after onset of GBS, presented as medians (inter-quartile range) and numbers ($n = 29$). The arrows show the change of scores for four patients showing some recovery during the time from 2 to 10 years after onset.

Median (IQR)	2 weeks 4 (2.5–5)	2 years 1 (1–1)	10 years 1 (0–1)
0 = healthy	0	10	10
1 = minor symptoms or signs, able to run	0	14	15
2 = able to walk more than 10 m without assistance but unable to run	7	1	1
3 = able to walk more than 10 m with assistance	4	1	2
4 = requires wheel-chair	10	3	1
5 = requiring assisted ventilation for at least part of the day	8	0	0
6 = dead due to Guillain–Barré syndrome	0	0	0

severe weakness was the lower part of the face, resulting in functional deficits such as inability to pucker the lips.

At 2 weeks after GBS all participants but one showed low BI scores, corresponding to dependency in ADL (Table 3). BI scores were stable on a group level from 1 to 10 years, with the same four participants (14%) showing a GBS-related dependency in ADL. Five additional participants also had decreased BI scores at 10 years; four persons were now in their mid eighties and had some difficulties handling stairs. One person had had poliomyelitis in childhood and this affected physical capacity. Thus, a total of nine of 29 participants (31%) had some dependency in ADL at the 10-year test occasion, however not due to GBS in all cases.

FAI scores increased slightly but significantly, between 1 year and 2 years; however, scores at 1 year and at 10 years did not differ significantly (Table 3).

An extensive impact on HRQL was seen at 2 weeks after onset of GBS, manifesting as high scores on the SIP, but scores had decreased significantly by the time of the other test occasions (Table 3 and Fig. 1). At 10 years, only the aggregated physical dimension mean score of the SIP was significantly higher than the reference data [28]. Conversely, scores in the emotional behavior category were significantly lower than reference data. In comparison to those who had fully recovered, the participants with GBS disability score > 1 had significantly higher mean scores in the four SIP categories body care and movement (14.0 vs. 0.9), home management (23.2 vs. 0), ambulation (19.6 vs. 2.3) and recreation and pastimes (26.3 vs. 4.9), but not in the other eight categories. The most marked item in the SIP at 10 years was ‘I walk more slowly’ ($n = 12$).

On the ONLS, six of 29 participants (21%) had affected functioning (i.e. score ≥ 2) in the upper body (Table 4), and six further participants had paresthesia at 10 years after onset of GBS, median score 0 (IQR 0–1). Fifteen (52%) were limited in activities such as climbing stairs and running as reflected in the ONLS lower body scores, median score 1 (IQR 0–1). Four participants had ONLS score ≥ 2 in both upper and lower bodies.

On the Walk-12, 23 participants (79%) rated limitations in walking such as walking more slowly or walking less smoothly due to residual symptoms of GBS at 10 years after onset (Table 5). Seventeen (59%) reported limited running ability. The median score of the Walk-12 was 23 (IQR 2–65). Participants with GBS disability score ≥ 1 had significantly higher mean scores on the Walk-12 than those who had fully recovered (48 vs. 9).

Table 3

Medians (inter-quartile ranges) of the measures of activity and health-related quality of life from onset of Guillain-Barré syndrome to 10 years afterwards (n = 29).

Measure (range of sum score)	2 weeks	1 year	2 years	10 years
Barthel Index (0–20)	6 (3–16)*	20 (20–20)	20 (20–20)	20 (20–20)
Frenchay Activity Index (0–45) ^a		31 (26–36)**	34 (28–38)	32 (26–36)
Sickness Impact Profile, total score (0–100)	40.4 (32.6–47.1)*	4.0 (0.4–12.4)	4.2 (0–10.4)	3.6 (1.4–11.4)
Sickness Impact Profile, physical dimension (0–100)	51.2 (42.3–58.3)*	3.1 (0–13.7)	3.8 (0–13.8)	5.1 (0–14.6)
Sickness Impact Profile, psychosocial dimension (0–100)	17.1 (11.2–31.7)*	1.4 (0–7.2)	1.9 (0–7.8)	2.0 (0–4.0)

^a Not measured at the 2 week test occasion.* Significantly different compared to scores at all later time-points ($p < 0.05$).** Significantly different ($p < 0.05$) compared to 2 years after onset but not 10 years.

The mean score of the FSS was 3.6 (SD 1.7). Participants with GBS disability score ≥ 1 had significantly higher mean scores on the FSS than those who had fully recovered (4.2 vs. 2.1).

3.3. Predicting long-term residual disability using the Erasmus GBS Outcome score

Median EGOS at 2 weeks (calculated retrospectively) was 4.5 (IQR 3.2–5.2) (Table 1). Nine participants were over 60 years of age at GBS onset and five had experienced diarrhea before onset. The GBS disability scores at 2 weeks are shown in Table 2. At 2 years after onset, there was only moderate significant correlation between the EGOS and scores on the BI ($r = 0.41$). At 10 years, significant high correlations were found between the EGOS at 2 weeks and the ONLS arm scale ($r = 0.70$). Significant but low correlations were found with the SIP categories body care and movement ($r = 0.39$) and communication ($r = 0.43$). There were no significant correlations between the EGOS and the other outcome measures at either 2 or 10 years after onset.

4. Discussion

In this prospective longitudinal study of a population-based cohort of persons with GBS, four of 29 participants (14%) had moderately to severely residual disability at 10 years after onset, defined as GBS disability score of ≥ 2 . A further 50% had minor symptoms such as paresthesia. The lower extremities were more affected than the upper ones, both at onset of GBS and at all subsequent evaluation time-points. Our previous studies [3,4] reported that functional recovery takes place mainly during the first 6–12 months after onset of GBS, although in the present study a small proportion (14%) of patients continued to improve slightly 2–10 years after onset. However, the five

participants with facial paralysis at 2 years showed no further recovery in facial function at 10 years, reflecting the poor prognosis of axonal damages to the facial nerve in GBS. This prospective study shows that the residual disabilities at 1–2 years after GBS onset largely continue to persist at 10 years post-onset. This information might help clinicians in providing GBS patients with information about the long-term prognosis, and also in the planning of rehabilitation services.

The residual disability assessed by our different scales at 1 year predicted long-term disability as few significant changes were found at 2–10 years. However, the EGOS managed only in part to predict residual disability at 2 and 10 years after onset, though this could be due to our small sample size. High correlation coefficients were only seen between the EGOS and the ONLS arm scale. In our sample, persistent disability in the arms was found in participants who were more severely affected in the whole body at all test occasions. No participant was more disabled in the arms than the legs. Further studies with larger study groups are needed to investigate the prognostic value of the EGOS for longer timescales than 6 months after onset of GBS.

The ONLS revealed disability of the upper body in six participants at 10 years after GBS onset and low FAI scores revealed corresponding limitations in doing household activities. These six participants had paresis in the arms from onset, with decreased hand grip strength [3]. At the earlier test occasions from onset to 2 years, neither the ONLS nor the ODSS were available and so disability of the upper body was not assessed. Functional status of the upper body should be included in the follow-up of patients with GBS, in order to discover whether orthotic devices and training may be needed.

Four of the participants showed slight improvements in GBS disability score between 2 and 10 years. Their recovery rates were

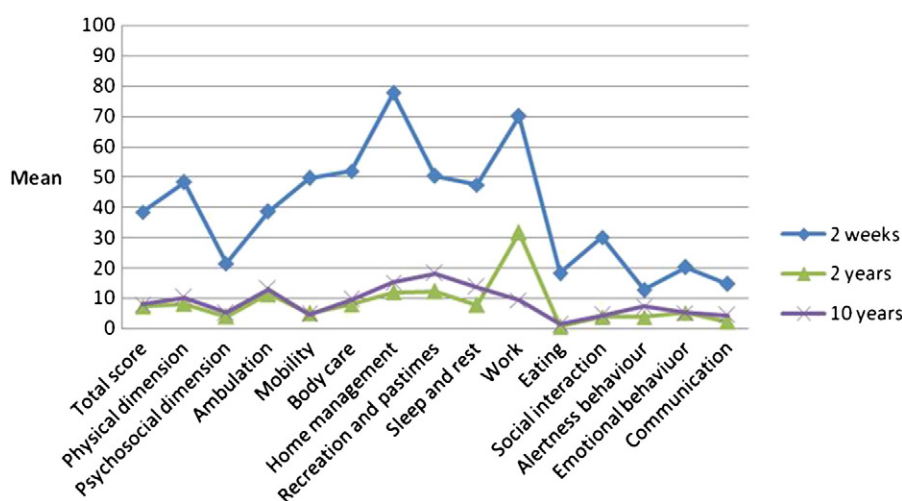


Fig. 1. Mean score on the Sickness Impact Profile at 2 weeks, 2 years and 10 years after onset of Guillain-Barré syndrome. The category 'work' does not include persons retired due to age.

Table 4

Distribution of scores on the Overall Neuropathy Limitations Scale at 10 years after onset of Guillain-Barré syndrome (n = 29).

Arm scale numbers	Leg scale numbers		
Normal	17	Walking/climbing stairs/running not affected	14
Minor symptoms	6	Walking/climbing stairs/running is affected but gait looks normal	10
Disability affecting function	4	Walks independently but gait looks abnormal	1
Disability preventing function	1	Requires unilateral support to walk 10 m	1
Purposeful movements possible	1	Requires bilateral support to walk 10 m	2
Purposeful movements prevented	0	Requires wheel-chair to travel 10 m	0
		Requires wheel-chair to travel any distance	1
		Purposeful movements prevented	0

slow, but persisted after 2 years. At the 10-year follow-up, all four reported that the recovery rate stopped at 3–4 years after onset. These four persons had typical GBS, with both motor and sensory symptoms, and continued with physical therapy for several years.

In the acute phase only three participants did not receive treatment with IVIG or plasma exchange. These participants had a mild GBS with symptoms below waist level, GBS disability scores of 2–3. During the first year they recovered and at 2 and 10 years had GBS disability scores of 0. Choice of IVIG or plasma exchange was made by the physician in charge at the participating hospitals. One participant was treated with plasma exchange plus corticosteroids.

Regaining the ability to walk is often a major goal of rehabilitation. Apart from timed objective measures, it is important to incorporate the patient's own perspective in the assessment. Even at 10 years after onset, about 40% of the participants reported limitations in walking speed, distance and ability to run on specific items on the Walk-12 (Table 5). However, only five had such decreased walking ability that assistive devices were needed. These results reflect the fact that residual symptoms may be subtle, and therefore measures of physical capacity need to include aspects such as longer walking distances. To prevent deterioration and facilitate physical activity, physical therapy may be needed for patients with residual symptoms. Although evidence is scarce, exercise seems to improve physical functioning [29,30].

Subjective dysfunction was more prominent in the physical dimension of the SIP than the psychosocial dimension at all test occasions. Notably lower scores on physical dimensions have also been found in cross-sectional studies of HRQL several years after GBS, using the SIP [31] and the SF-36 [14,32]. Thus, it seems that persons with GBS recover quite well in terms of the psychosocial aspects of HRQL. Participants with residual physical symptoms reported decreased HRQL in the SIP categories that required walking capacity,

balance and use of hands. The findings of this study support the importance of including both objective and subjective measures in the long-term follow-up after GBS.

At the test occasions at 1 and 2 years, 40% of the patients rated their fatigue using a visual analogue scale [3]. In the present study, we decided to use the FSS instead as it is a well-known measure of fatigue. Earlier, Merkies et al. [26] found that 80% of patients with residual symptoms 3–6 years after onset of GBS experienced fatigue. A large postal survey in England [12] found similar results to those in the present study, namely that fatigue seems to be more severe in persons with residual symptoms.

A limitation of this study is the relatively small study group (n = 29). However, at onset of GBS, the clinical characteristics of our sample were similar to those reported in other larger population-based GBS studies [5,7,8,33]. The reliability of the data is strengthened by the fact that most of the data collection over the 10 years was performed by the same researcher (A.F.). In addition, during the home visits, the researcher confirmed the participants' ability to adequately rate their physical capacity on the self-rating questionnaires.

In conclusion, this prospective study covering the 10 years after GBS onset confirms that the residual disability seen at 1–2 years after onset is probably life-long. Physical disability at 10 years after GBS onset was associated with self-reported significant fatigue, higher subjective dysfunction in physical HRQL and higher impact on walking ability. The EGOS calculated in the acute phase of GBS could partially predict the extent of residual disability at 2 years and 10 years.

Conflicts of interest

There are no competing interests.

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Table 5

Self-rated limitations in walking assessed with the Walk-12 at 10 years after onset of Guillain-Barré syndrome (n = 29). Number of persons.

In the past 2 weeks how much have residual symptoms after GBS...	Not at all	A little/moderately	Quite a bit/extremely
Limited your ability to walk?	12	5	12
Limited your ability to run?	12	3	14
Limited your ability to climb stairs?	12	9	8
Made standing more difficult?	20	6	3
Limited your balance?	16	7	6
Limited your walking distance?	13	5	11
Increased the effort needed for walking?	11	8	10
Made it necessary to use support indoors?	20	4	5
Made it necessary to use support outdoors?	20	4	5
Slowed down your walking?	13	4	12
Affected how smoothly you walk?	9	11	9
Made you concentrate on your walking?	14	5	10

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